USN | | | | | | | | | | [22TCSE256]

FIFTH SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

DISASTER RECOVERY

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	What is disaster recovery in the context of IT systems?	L1	CO1	**02**
	b.	What is a natural disaster?	L1	CO2	**02**
	c.	What are seismic zones, and why are they important in disaster recovery planning for cloud infrastructure?	L1	CO3	**02**
	d.	What is disaster prevention?	L1	CO4	**02**
	e.	What is the scope of disaster management?	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	A company experiences a ransomware attack. Outline the steps the disaster recovery team should take to restore critical services.	L3	CO1	**05**
		OR			
	b.	What roles and responsibilities would you assign to the team members when tasked with building a disaster recovery team?	L3	CO1	**05**
3.	**a.**	During a nuclear reactor meltdown, outline the immediate safety measures that should be taken to minimize human exposure to radiation.	L3	CO2	**05**
		OR			
	b.	Compare the effects of man-made disasters like industrial accidents with natural disasters in terms of environmental damage.	L 4	CO2	**05**
4.	**a.**	Explain the difference between cloud infrastructure regions and availability zones in the context of disaster recovery.	L 4	CO3	**05**
		OR			
	b.	Let's design a disaster recovery solution for a cloud-based application. The data centers exist within a region which is predisposed to cyclones, how would you architect it towards achieving high availability and having good disaster recovery?	L 4	CO3	**05**

5.	**a.**	Design a preparedness plan for responding to a biological disaster such as a pandemic in a densely populated city.	L 3	CO4	**05**
		OR			
	b.	Analyze the effectiveness of early warning systems in reducing the impact of natural disasters.	L 4	CO4	**05**
6.	**a.**	Design a mitigation strategy to prevent fires in urban areas.	L 3	CO5	**05**
		OR			
	b.	Evaluate the effectiveness of disaster mitigation strategies in preventing large-scale loss during an earthquake in a densely populated city.	L 4	CO5	**05**
7.	**a.**	How can meteorological data be used in disaster risk evaluation for hurricanes?	L 3	CO6	**05**
		OR			
	b.	Analyze the differences between structural and non-structural mitigation strategies.	L 4	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Design a disaster recovery plan for an organization that primarily operates through cloud services. What key components should be included in the plan?	L 5	CO1	**10**
		OR			
	b.	Evaluate the effectiveness of using cloud-based disaster recovery compared to traditional on-premise disaster recovery strategies.	L 5	CO2	**10**
9.	**a.**	Design an emergency response plan for a coastal city facing an approaching hurricane, focusing on evacuation, communication, and resource allocation.	L 5	CO2	**10**
		OR			
	b.	Create a public awareness campaign aimed at educating people about the risks and preparedness measures for a potential industrial accident in a community near a chemical plant.	L 5	CO1	**10**
10.	**a.**	Analyze the potential risks associated with placing a cloud data center in a seismic zone and how these risks can be mitigated in a disaster recovery strategy.	L 4	CO2	**10**

		OR			
	b.	Design a disaster recovery plan for a cloud infrastructure that operates in a region with both seismic activity and frequent cyclones. Include considerations for both physical and cloud-based redundancies.	L 5	CO2	**10**
11.	**a.**	Create a disaster mitigation strategy to reduce the impact of flooding in a coastal city.	L 5	CO3	**10**
		OR			
	b.	Evaluate the role of risk assessment in disaster preparedness, especially in regions prone to biological disasters.	L 6	CO4	**10**
12.	**a.**	Create a comprehensive disaster mitigation program for a coastal city prone to hurricanes and floods.	L 5	CO3	**10**
		OR			
	b.	Evaluate the role of emerging technologies, such as remote sensing and AI, in enhancing disaster mitigation strategies.x	L 5	CO5	**10**
13.	**a.**	Critically assess the importance of testing a disaster recovery plan regularly. What risks are associated with an untested disaster recovery plan?	L 6	CO4	**10**
		OR			
	b.	Evaluate the advantages and disadvantages of using cloud-based disaster recovery solutions compared to traditional on-premise disaster recovery in regions prone to natural disasters.	L 6	CO6	**10**